Joshua Liu

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EDUCATION

TUFTS UNIVERSITY
M.S. in Artificial Intelligence
Expected May 2028 | Medford, MA

BRANDEIS UNIVERSITY
B.S. in CS & Applied Math
Magna Cum Laude
Phi Beta Kappa | GPA: 3.92
Grad. May 2026 | Waltham, MA

LINKS

Github:// **joshliu14**
LinkedIn:// **joshuagliu**
YouTube:// **joshualiu14**

COURSEWORK

UNDERGRADUATE

Data Structures and Algorithms
Advanced Algorithms
Operating Systems
Computer System Design
Embedded Systems
Intro to Machine Learning
Database Management Systems
Fundamentals of SWE
Web Application Development
Theory of Computation
Big Data and Numerical Methods

SKILLS

PROGRAMMING

Advanced: Python • Java • JavaScript
SQL • C++ • TypeScript
Proficient: Matlab • LaTeX • Lua
Swift • Go • CSS • HTML

FRAMEWORKS

PyTorch • TensorFlow • OpenCV
React.js • Node.js • Electron.js
Flask • FastAPI • MongoDB

TOOLS/PLATFORMS

Git • GitHub Actions • Docker
Azure • GCP • AWS • Firebase
Linux • Vim • Anaconda

OTHER

TEACHING

TA, COSI 12B (Java)
BUGS Tutor: Python, Java, DSA, OS

VARSITY ATHLETE

Brandeis University Baseball

EXPERIENCE

GLAUKOS | Applied Research Co-op
July 2025 – Dec. 2025 | Burlington, MA

- Leveraged Meta's SAM2 to automate ground-truth generation and trained optimized YOLO models, improving F1 score by 51%.
- Built an Electron.js application for image labeling and automated event-based data sorting, increasing data-processing throughput by 94% across 300+ patient datasets (100+ clinical).
- Eliminated race conditions and optimized vision software via multithreading and concurrency controls (29% increase in FPS), and automated calibration in Python.

THE COMMONS XR | Data Engineer and DevOps Intern
June 2024 – Sept. 2024 | San Diego, CA

- Led daily standups; built GitHub Actions CI workflows that automatically run the test suite on every pull request, eliminating any manual testing per merge.
- Automated the post-session analytics pipeline with Azure Logic Apps, cutting session-to-dashboard time from a manual refresh to under 1 minute in Power BI.
- Designed and tested a new database schema to improve relationship modeling and integrate additional software-generated data.

BOSTON COLLEGE (CV DEPT.) | Full Stack Engineer Intern
June 2023 – Aug. 2023 | Chestnut Hill, MA

- Built features for a neuron segmentation and labeling **web app** (React.js, FastAPI).
- Developed a multi-tab view running concurrent **Neuroglancer** instances, letting researchers compare neuron data side by side without switching contexts.
- Built temporary and permanent file labeling for high-resolution neuron images.

RESEARCH/PROJECTS

TUFTS DATACENTER SYSTEMS LAB | Graduate Researcher
Sept. 2026 – Present | Medford, MA

- Profiling CPU inference of machine learning models with `perf` to identify memory- and compute-bound bottlenecks and underutilized cores.
- Developing optimizations to maximize CPU utilization and reduce inference latency.

CORE ML LAB & BRAVA LAB | Undergrad Researcher
June 2024 – May 2025 | Waltham, MA

- Core Machine Learning Lab (**Hongfu Liu**): Researched improving the computational efficiency of wrapper methods using Hessian-free influence functions.
- BraVA Lab (**Dylan Cashman**): Conducted grounded theory qualitative research and built an LLM-integrated student career support tool (React.js, Google Gemini).

MY GENIUS
Jan. 2025 – Apr. 2025

- Implemented an AI chat feature and document upload using the OpenAI API; fixed backend routing for exam sharing links and redesigned the dashboard UI.
- Deployed and managed the web app on AWS Elastic Beanstalk.

PROSCOUT AI
Sept. 2024 – Dec. 2024

- Built real-time OpenCV image analysis and a React dashboard for camera control.
- Designed scalable cloud infrastructure on Google Cloud Platform (GCP).

TRAVELMUG
Dec. 2023 – May 2024

- Built a cafe recommender (React.js, Redux, Firebase) with Docker and Azure CI/CD.